

HDOC Day-18 Report

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Batch: 15

Menu-Driven C Program for Digit-Based Operations

The program uses switch-case and separate user-defined void functions to check Strong, Armstrong, Harshad and Duck numbers.

Problem Statement

Write a menu-driven C program using **switch-case** to perform digit-based operations on a positive integer. Define one separate user-defined function for each menu option, with **void** return type and a simple function call. Each function reads the required input, processes it, and displays the result.

Operations

1. Check whether the number is a Strong number.
2. Check whether it is an Armstrong number.
3. Check whether it is a Harshad number.
4. Check whether it is a Duck number.

Algorithm

Main Program:

1. Start.
2. Display four menu choices.
3. Read the user's choice.
4. Use switch-case to call the corresponding void function.
5. Display an invalid-choice message for any other option.
6. Stop.

Strong Number: Read n, extract every digit, calculate its factorial, add the factorials, and compare the sum with n.

Armstrong Number: Read n, extract every digit, add the cube of each digit, and compare the sum with n.

Harshad Number: Read n, calculate the sum of its digits, and check whether n is divisible by that sum.

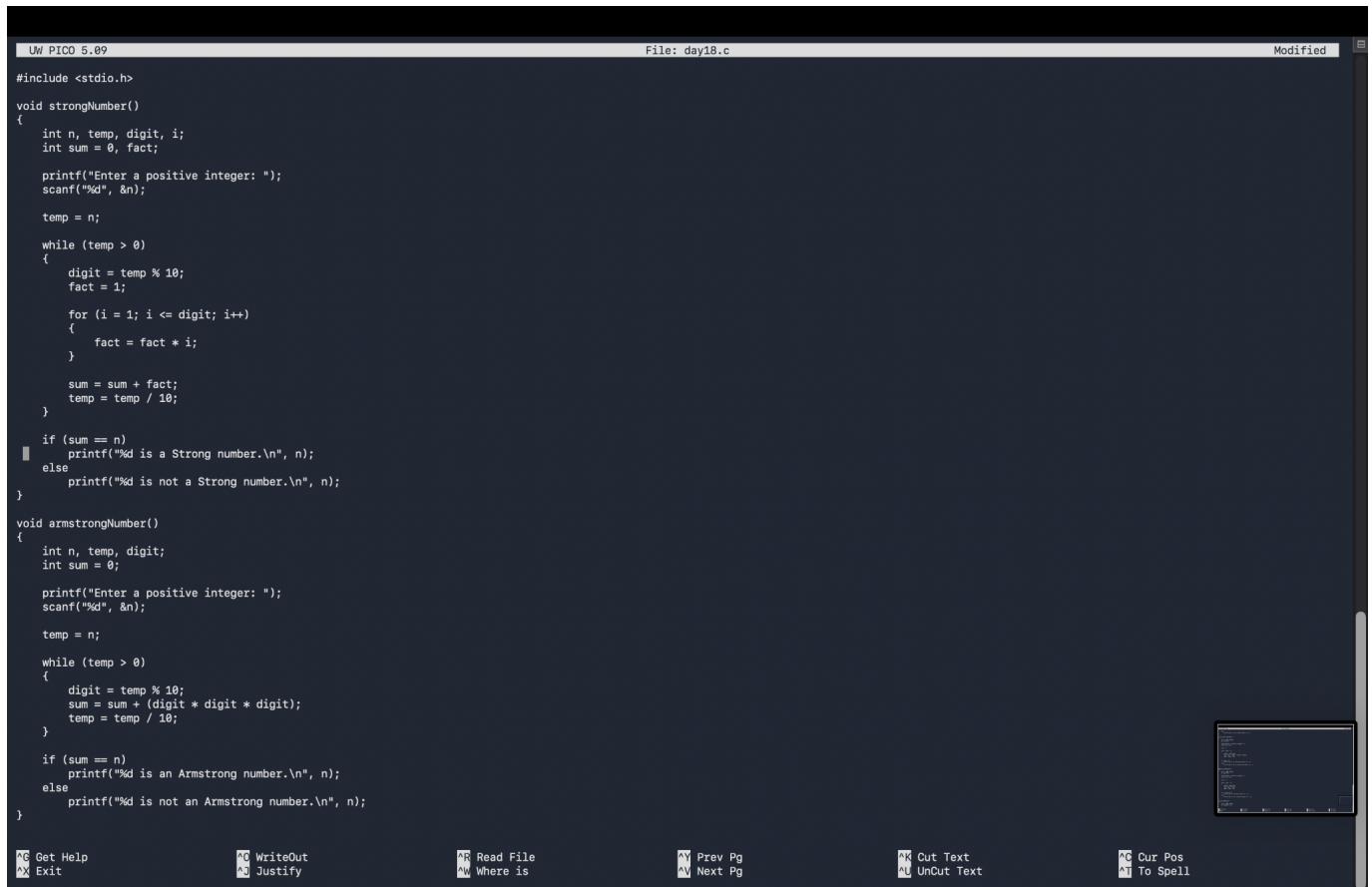
Duck Number: Read n, examine its digits, and check whether at least one digit is 0.

Test Case Table

Test Case	Menu/Input	Expected Output	Result
1	1, 145	145 is a Strong number.	Pass
2	2, 153	153 is an Armstrong number.	Pass
3	3, 18	18 is a Harshad number.	Pass
4	4, 102	102 is a Duck number.	Pass

Program Screenshots

The following screenshots show the Day-18 C program written with separate functions and a switch-case menu.



The screenshot shows a C program in a text editor. The editor has a title bar with 'UW PICO 5.09', 'File: day18.c', and 'Modified'. The code defines two functions: `strongNumber()` and `armstrongNumber()`. Both functions take an integer `n` and calculate the sum of its digits raised to the power of the number of digits. The `strongNumber()` function checks if the sum equals `n`, and the `armstrongNumber()` function checks if the sum equals `n`. Both functions print the result and return. A small window titled 'day18.c' is visible in the bottom right corner of the editor.

```
#include <stdio.h>

void strongNumber()
{
    int n, temp, digit, i;
    int sum = 0, fact;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    temp = n;

    while (temp > 0)
    {
        digit = temp % 10;
        fact = 1;

        for (i = 1; i <= digit; i++)
        {
            fact = fact * i;
        }

        sum = sum + fact;
        temp = temp / 10;
    }

    if (sum == n)
        printf("%d is a Strong number.\n", n);
    else
        printf("%d is not a Strong number.\n", n);
}

void armstrongNumber()
{
    int n, temp, digit;
    int sum = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    temp = n;

    while (temp > 0)
    {
        digit = temp % 10;
        sum = sum + (digit * digit * digit);
        temp = temp / 10;
    }

    if (sum == n)
        printf("%d is an Armstrong number.\n", n);
    else
        printf("%d is not an Armstrong number.\n", n);
}
```

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```
    else
        printf("%d is not a Strong number.\n", n);
}

void armstrongNumber()
{
    int n, temp, digit;
    int sum = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    temp = n;

    while (temp > 0)
    {
        digit = temp % 10;
        sum = sum + (digit * digit * digit);
        temp = temp / 10;
    }

    if (sum == n)
        printf("%d is an Armstrong number.\n", n);
    else
        printf("%d is not an Armstrong number.\n", n);
}

void harshadNumber()
{
    int n, temp, digit;
    int sum = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    temp = n;

    while (temp > 0)
    {
        digit = temp % 10;
        sum = sum + digit;
        temp = temp / 10;
    }

    if (n % sum == 0)
        printf("%d is a Harshad number.\n", n);
    else
        printf("%d is not a Harshad number.\n", n);
}

void duckNumber()
{
    int n, temp, digit;
    int hasZero = 0;

```

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```
{
    int n, temp, digit;
    int sum = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    temp = n;

    while (temp > 0)
    {
        digit = temp % 10;
        sum = sum + digit;
        temp = temp / 10;
    }

    if (n % sum == 0)
        printf("%d is a Harshad number.\n", n);
    else
        printf("%d is not a Harshad number.\n", n);
}

void duckNumber()
{
    int n, temp, digit;
    int hasZero = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    temp = n;

    while (temp > 0)
    {
        digit = temp % 10;

        if (digit == 0)
        {
            hasZero = 1;
            break;
        }

        temp = temp / 10;
    }

    if (hasZero)
        printf("%d is a Duck number.\n", n);
    else
        printf("%d is not a Duck number.\n", n);
}

int main()
{
    int choice;
```

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```
scanf("%d", &n);

temp = n;

while (temp > 0)
{
    digit = temp % 10;

    if (digit == 0)
    {
        hasZero = 1;
        break;
    }

    temp = temp / 10;
}

if (hasZero)
    printf("%d is a Duck number.\n", n);
else
    printf("%d is not a Duck number.\n", n);
}

int main()
{
    int choice;

    printf("----- Digit Based Operations ----- \n");
    printf("1. Check Strong Number\n");
    printf("2. Check Armstrong Number\n");
    printf("3. Check Harshad Number\n");
    printf("4. Check Duck Number\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            strongNumber();
            break;

        case 2:
            armstrongNumber();
            break;

        case 3:
            harshadNumber();
            break;

        case 4:
            duckNumber();
            break;

        default:
    }
```

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```
printf("1. Check Strong Number\n");
printf("2. Check Armstrong Number\n");
printf("3. Check Harshad Number\n");
printf("4. Check Duck Number\n");

printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice)
{
    case 1:
        strongNumber();
        break;

    case 2:
        armstrongNumber();
        break;

    case 3:
        harshadNumber();
        break;

    case 4:
        duckNumber();
        break;

    default:
        printf("Invalid choice.\n");
}

return 0;
}
```

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Execution and Result

The program should be compiled and executed after saving the Day-18 source file. Test each menu option using suitable positive integers such as 145, 153, 18 and 102.

Note: The execution screenshot supplied shows an older cylinder program rather than the Day-18 menu-driven program, so it has not been presented as the Day-18 execution result.

Conclusion

The menu-driven program demonstrates switch-case selection, user-defined void functions, loops, conditional statements and digit-based calculations in C.